

No. of Functional Features Included in the Solution

Date	30 June 2026
Team ID	XXXXXX
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	User Interface	Provides an interactive web interface for users to enter HDI prediction.	Frontend (HTML, CSS)	Done	High
2	Dataset Loading	Reads the HDI.csv dataset and loads it into the application for preprocessing.	Data Processing	Done	High
3	Data Preprocessing	Handles missing values, checks data quality, and prepares data for model training.	Machine Learning	Done	High
4	Data Visualization	Generates strip plots, correlation heatmaps, and other visualizations for data analysis.	Visualization	Done	Medium
5	HDI Prediction Model	Trains a Linear Regression model to predict Human Development Index values.	Machine Learning	Done	High
6	Model Saving	Saves the trained model as HDI.pkl for deployment.	Model Storage	Done	Medium
7	Flask Web Application	Connects the trained model with the web interface and processes prediction requests.	Backend (Flask)	Done	High
8	HDI Prediction Result	Displays the predicted HDI score and corresponding category (Low, Medium, High, Very High).	Prediction Module	Done	High

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8

Core / Must-Have Features	8
Additional / Nice-to-Have Features	2
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Home Page, Prediction Page
2	Backend / Logic	3	Flask Backend, Linear Regression Model, Prediction Logic
3	Database / Storage	2	HDI.csv Dataset, HDI.pkl Model
4	API / Integration	1	Flask Routing Between Frontend and Backend
5	Security / Authentication	1	Input Validation and Error Handling